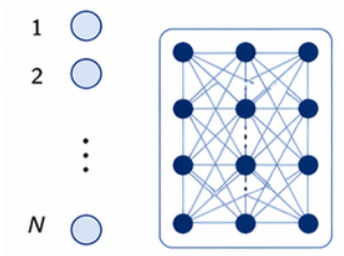
V1 Dense (deterministic)

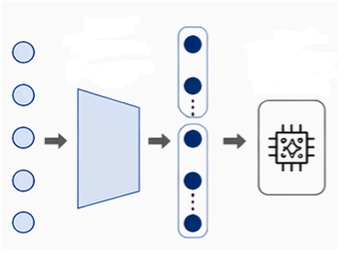


Dense relations, deterministic dimensionality reduction.

$\mathcal{I}_{V1} = \{(i,j) \mid 1 \le i,j \le q\}$

W_T: not used W_S: not used θ: trainable

V2 Dense (learned projection)

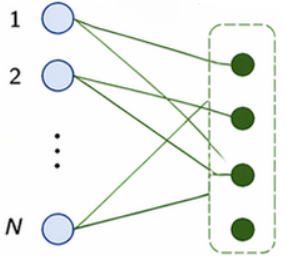


Dense relations, learned linear projection.

$\mathcal{I}_{V2} = \mathcal{I}_{V1}$

W_T: fixed W_S: trainable θ: trainable

V3 Anchor

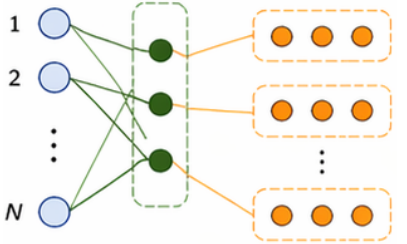


Sparse relations: computed only among anchor samples; other samples do not contribute.

$\mathcal{I}_{V3} = \{(a,a') \mid a,a' \in \mathcal{A}\}, |\mathcal{A}| = 8$

W_T: fixed W_S: trainable θ: trainable

V4 Anchor + kNN

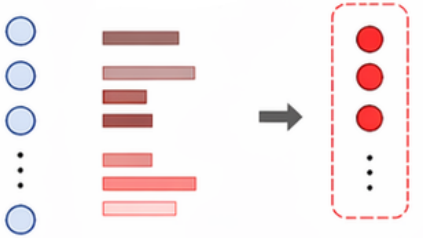


Adds local kNN relations for every sample, alongside anchor-to-anchor relations.

$\mathcal{I}_{V4} = \mathcal{I}_{V3} \cup \{(i,j) \mid j \in \mathcal{N}_k(i)\}, k = 4$

W_T: fixed W_S: trainable θ: trainable

V5 Guided



Dense relations, but restricted to the subset of samples selected by the teacher.

$\mathcal{I}_{V5} = \{(i,j) \mid i,j \in \mathcal{S}\}$, top-$q$ by teacher score

W_T: fixed W_S: trainable θ: trainable